

Reinforced Cement Concrete Structures

According to (IS 456-2000)

1) Properties of Concrete

- 1) Minimum Grade of Concrete for R.C.C = M20 (in Normal)
= M30 (in Sea water)
- 2) Unit weight of R.C.C. = 25 kN/m^3

3) Characteristic Compressive Strength (f_{ck})

We conduct various tests/specimens for various concrete to calculate f_{ck} .

Hence for design purpose we fix certain f_{ck} value by using formula.

(It gives 95% of values are greater than this value, i.e. only 5% are expected to fall below that value).

$$f_{ck} = f_{cm} - 1.65\sigma$$

$$\sigma = \text{Standard deviation} = \sqrt{\frac{\sum (F_i - F_m)^2}{n-1}}$$

F_i = load for each test, F_m = Avg load of all tests
 n = No. of tests

$$f_{cm} = \text{Mean strength} = \frac{\sum F_i}{n}$$

$$\sigma = 3.5 \text{ N/mm}^2 \text{ for (M10 - M15)}$$

$$= 4.0 \text{ N/mm}^2 \text{ (M20 - M25)}$$

$$= 5.0 \text{ N/mm}^2 \text{ (} \geq \text{M30)}$$

4) Flexural and Split Tensile Strength (f_{cr})

$$f_{cr} = 0.7 \sqrt{f_{ck}}$$

5) Modulus of Elasticity (E_c)

Also called 'Initial tangent modulus' or 'Short term elastic modulus'.

$$E_c = 5000 \sqrt{f_{ck}}$$

6) Shrinkage (Volumetric) Strain in Concrete

$$= 0.0003$$

(Three zero's after.)

7) Creep of Concrete (θ)

Is defined as the increase in deformation with time under constant load.

$$\text{Creep Coefficient } (\theta) = \frac{\text{Ultimate Creep strain}}{\text{Elastic strain at the case of loading}}$$

$$\theta = 2.0 \text{ (7 days)}$$

$$= 1.6 \text{ (28 days)}$$

$$= 1.1 \text{ (1 year)}$$

$$\therefore \text{Long term modulus of Elasticity} = \frac{\text{Short term Elasticity}}{1+\theta} = \frac{E_c}{1+\theta}$$

8) Classification of Concrete for R.C.C

1. Ordinary Concrete

M10 - M20

2. Standard Concrete

M25 - M55

3. High strength Concrete

> M60

(Max Cement Content = 450 kg/m^3)

2) Properties of Steel

$E_s = 2 \times 10^5 \text{ N/mm}^2$ (for all type of Reinforcement steel)

- * $\phi \leq 12 \text{ mm}$ - termed as bars & can be coiled for transportation
- $> 12 \text{ mm}$, termed as rods & they are transported in lengths.

* The size of bars used as Reinforcement are

5, 6, 8, 10, 12, 16, 20, 22, 25, 32, 36, 40, 45, 50 mm.

* Fe-250 (Mild or Medium Tensile Steel bars)

Fe415 & Fe500 (Highly Yield Deformed Cold worked Steel bars)

i.e. HYSD

Working Stress (or) Elastic method

The working stress method results in comparatively large sections.

→ Hence the failure of structure occurs at much higher loads

→ It does not provide realistic measure of F.O.S against collapse of structure.

Assumptions in W.S.M

→ Plane sections before and after bending remain plane.

→ All tensile stresses are taken up by reinforcement only

→ The stress-strain relation for steel and concrete is straight-line.

Permissible stress in concrete

The stresses in steel & concrete are not allowed to exceed some specified values of stresses are known as permissible stresses.

$$\therefore \text{Permissible stress in concrete} = \frac{\text{Characteristic strength (} f_{ck} \text{)}}{\text{Factor of safety for concrete}}$$

Nature of Beam in	F.O.S	Permissible stress in	
1. Flexure (Bending compression)	3	σ_{cbc}	$\approx f_{ck}/3$
2. Direct Compression	4	σ_{cc}	$\approx f_{ck}/4$
3. Bending Tension	14	σ_{tb}	$\approx f_{ck}/14$
4. Direct Tension	8.5-9.5	σ_{td}	$\approx f_{ck}/8.5 \sim f_{ck}/9.5$
5. Average Bond Shear	25-35	τ_{bd}	$\approx f_{ck}/25 \sim f_{ck}/35$

Permissible stress in steel

$$\therefore \text{Permissible tensile stress in steel} = (\sigma_{st})$$

$$\frac{\text{Yield strength (} f_y \text{)}}{\text{F.O.S. for steel}}$$

$$\text{F.O.S for steel reinforcement} = 1.8$$

$$\Rightarrow \sigma_{st} = \frac{f_y}{1.8}$$

Type of steel	σ_{st} (N/mm ²)
Fe 250	140 ; $\phi \leq 20\text{mm}$ 130 ; $\phi > 20\text{mm}$
Fe 415	230
Fe 500	275

Grade of concrete	σ_{cbc} (N/mm ²)
M15	5
M20	7
M25	8.5
M30	10

$$\therefore \text{Strain in concrete } \epsilon_c = \frac{\sigma_{cbc}}{E_c} = \frac{f_c}{E_c}$$

$$\text{Strain in Reinforcement } \epsilon_s = \frac{\sigma_{st}}{E_s} = \frac{f_{st}}{E_s}$$

At bottom fibre ($\epsilon_c = \epsilon_s$)

$$\Rightarrow f_{cb} = \frac{f_{st}}{(E_s/E_c)} = \frac{f_{st}}{m}$$

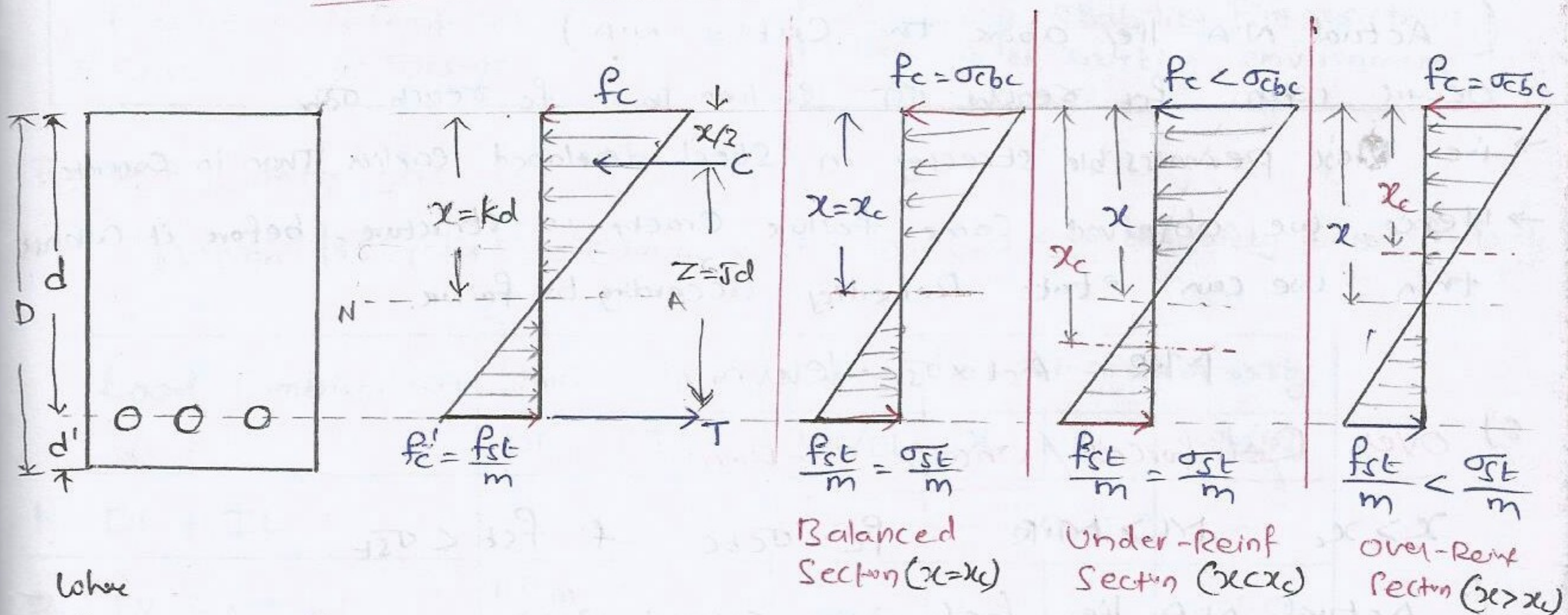
Modular Ratio (m)

$$m = \frac{E_s}{E_c} \quad (\text{if Temp \& Shrinkage effects are not taken into Consideration})$$

$$= \frac{280}{3\sigma_{cbc}} \quad (\text{if Temp \& Shrinkage effects are taken into Consideration})$$

	M15	M20	M25	M30
m	18.67	13.33	11	9.33

Singly Reinforced Sections



Where

f_c = Stress (Compressive) in concrete at top most fibre

$f'_c = \frac{f_{st}}{m}$ = Stress (Tensile) in concrete at bottom most fibre

$x = kd$ = Depth of actual N.A (Passing through zero stress of $\frac{f_{st}}{m}$ & f_c)

$x_c = k_c d$ = Depth of Critical N.A (Passing through zero permissible stress of $\frac{\sigma_{st}}{m}$ & σ_{cbc})

$z = jd$ = lever arm = (Distance from C.G. of Compression zone - bottom fibre)

$$\text{N.A Coefficient (K)} = \frac{1}{1 + \frac{f_{st}}{m f_c}} \quad \& \quad k_c = \frac{1}{1 + \frac{\sigma_{st}}{m \sigma_{cbc}}} = \frac{280}{280 + 3 \sigma_{cbc}}$$

$$\text{Lever arm Coefficient (J)} = \left(1 - \frac{k}{3}\right) \Rightarrow \text{lever arm (Z)} = d - \frac{kd}{3}$$

Forces

Compressive force (C) = Area of compressive stress diagram $\times f_c = \left(\frac{1}{2} b x\right) f_c$

Tensile force (T) = Area of tension steel $\times f_{st} = A_{st} \times f_{st}$

Location of N.A

From static equilibrium, Compressive force (C) = Tensile force (T) $\Rightarrow \frac{1}{2} b x f_c = A_{st} f_{st}$

Moment of Resistance

$$M.R = C \times \text{lever arm} = T \times \text{lever arm}$$

a) Balanced/Economical/Critical Section

$$x = x_c \quad M = M.R \quad f_c = \sigma_{cbc} \quad f_{st} = \sigma_{st}$$

occurs when "Permissible stresses in steel and concrete are developed simultaneously". (i.e. concrete will crush & steel will yield at same time)

$$\therefore \boxed{M.R = Q b d^2} \quad \left(\because M.R = C \times z = \frac{1}{2} b (k_d) \sigma_{cbc} \times (j_d) \right. \\ \left. = \left(\frac{1}{2} \sigma_{cbc} j_d \right) b d^2 \quad \therefore Q = \frac{1}{2} \sigma_{cbc} j_d \right)$$

$$\text{or} \\ M.R = A_{st} \sigma_{st} \times \text{lever arm}$$

b) Under Reinforced/Safe Section

$$x < x_c \quad M < M.R \quad f_c < \sigma_{cbc} \quad \& \quad f_{st} = \sigma_{st}$$

($\because$ Actual N.A. lies above the Critical N.A.)

occurs when f_{st} reaches σ_{st} earlier than f_c reaches σ_{cbc}

$\rightarrow$ i.e. Max permissible stresses in steel developed earlier than in concrete.

$\rightarrow$ Hence, we observed some failure cracks in structure, before it collapses then we can take remedy according to failure.

$$M.R = A_{st} \times \sigma_{st} \times \text{lever arm}$$

c) Over Reinforced/Unsafe Section

$$x > x_c \quad M > M.R \quad f_c = \sigma_{cbc} \quad \& \quad f_{st} < \sigma_{st}$$

($\because$ Actual N.A. lies below the Critical N.A.)

occurs when f_c reaches σ_{cbc} earlier than f_{st} reaches σ_{st}

$\rightarrow$ i.e. Max permissible stresses in concrete developed earlier than in steel.

$\rightarrow$ Hence, we observe cannot observe any failure cracks in structure before it collapses. And it is suddenly collapsed.

$$M.R = Q b d^2$$

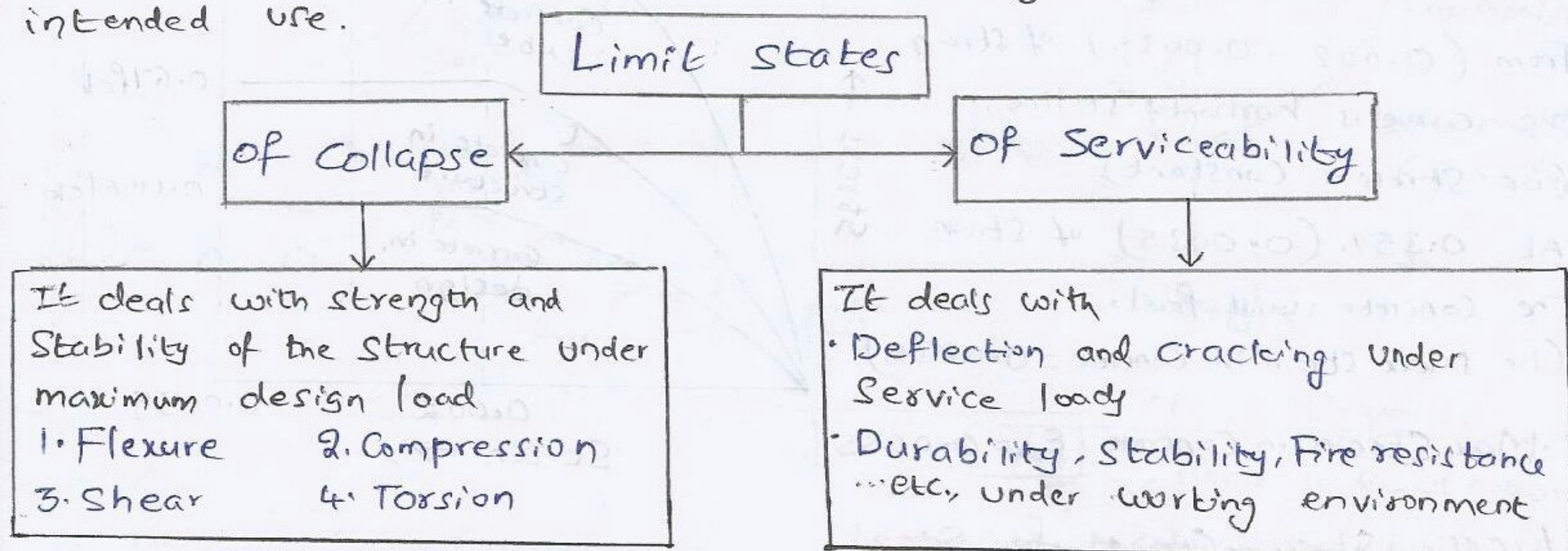
Note:

The concrete below the N.A. does not resist any bending moment.

But simply serve to embed the tensile steel.

Limit State (or) Plastic Method

In limit state method, a structure is designed for safety against collapse (i.e. for ultimate strength to resist ultimate load) and checked for its serviceability at working loads, thus rendering the structure fit for intended use.



Partial Safety factors for loads

$$\text{Design load (F}_d\text{)} = \text{Characteristic load (F)} \times \text{partial safety factor for loads (}\gamma_f\text{)}$$

Load Combination	Limit state of Collapse			Limit state of Serviceability		
	DL	IL	WL/EL	DL	IL	WL/EL
1. DL + IL	← 1.5 →			← 1.0 →		
2. DL + WL/EL	1.5 0.9	-	1.5	1.0	-	1.0
3. DL + IL + WL/EL	← 1.2 →			1.0	← 0.8 →	

Note

- 0.9 value to be consider when stability against overturning / stress reversal is consider
- If we consider the EL in design, then there is no need to consider WL, b'c EL is already too high. (i.e. combination of WL + EL is not consider)

Partial Safety factors for material

$$\text{Design strength of material (F}_d\text{)} = \frac{\text{Characteristic strength of material (F}_{ck}\text{)}}{\text{Partial safety factor for material (}\gamma_m\text{)}}$$

Limit State of Collapse	Limit State of Serviceability
Steel - 1.15	Steel 1.0
Concrete - 1.50	Concrete 1.0

Steel

$$\text{Design strength of steel} = \frac{f_y}{1.15} = 0.87 f_y$$

Concrete

f_{ck} is consider for 150mm cube. But in actual the size will vary of high. Hence for design purpose consider $f_{ck} = \frac{2}{3} f_{ck} = 0.67 f_{ck}$ ($\because f_{ck} \propto \frac{1}{\text{size}}$)

$$\text{Design strength of Concrete} = \frac{\frac{2}{3} f_{ck}}{1.5} = \frac{0.67 f_{ck}}{1.5} = 0.446 f_{ck}$$

Stress-Strain Curve for Concrete

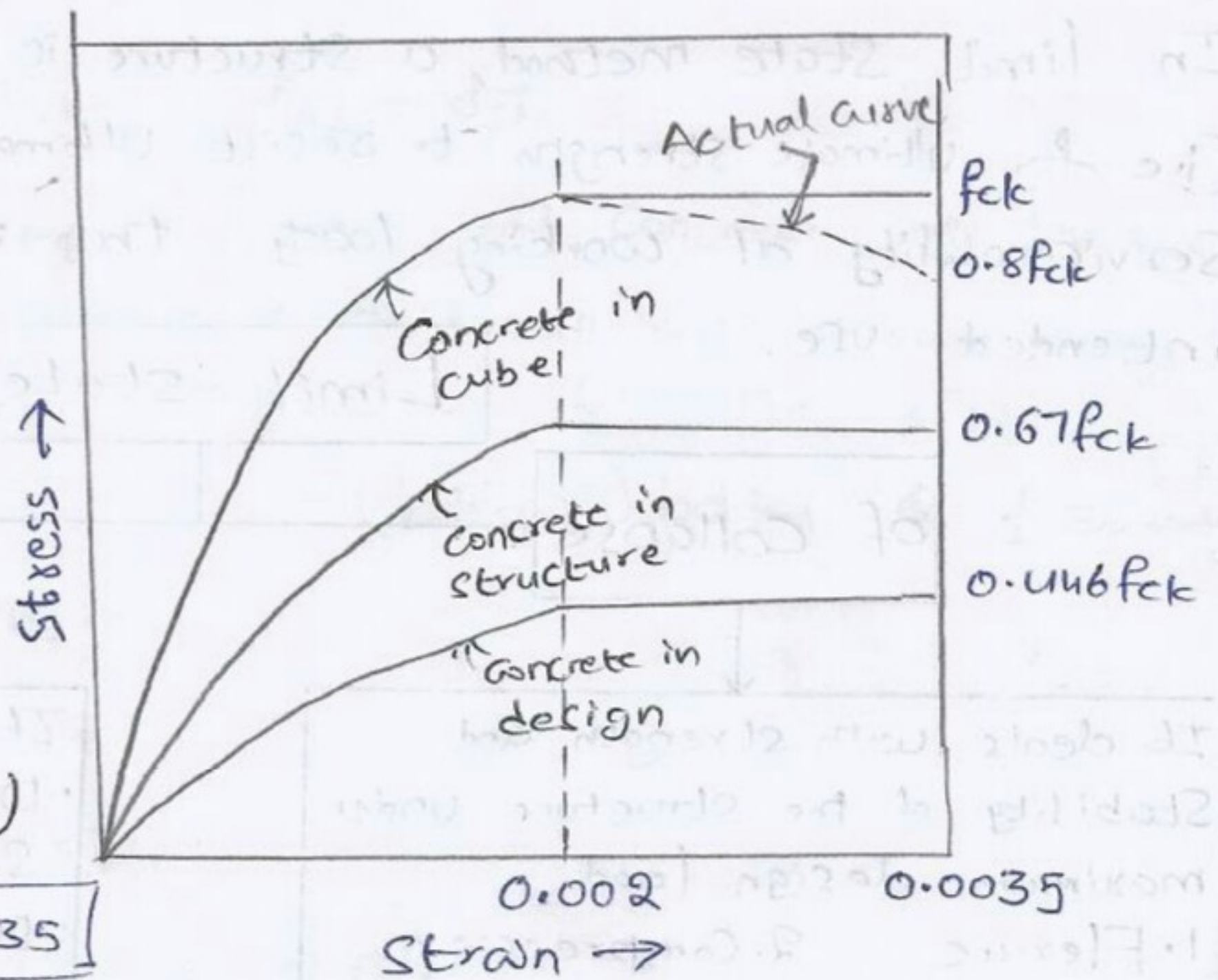
→ upto 0.2% (0.002) of strain

the curve is parabolic
(i.e. Stress & Strain)

→ From (0.002 - 0.0035) of strain
the curve is horizontal straight line.
(i.e. Stress is Constant)

→ At 0.35% (0.0035) of strain
the concrete will fail.
(i.e. Max strain in concrete = 0.0035)

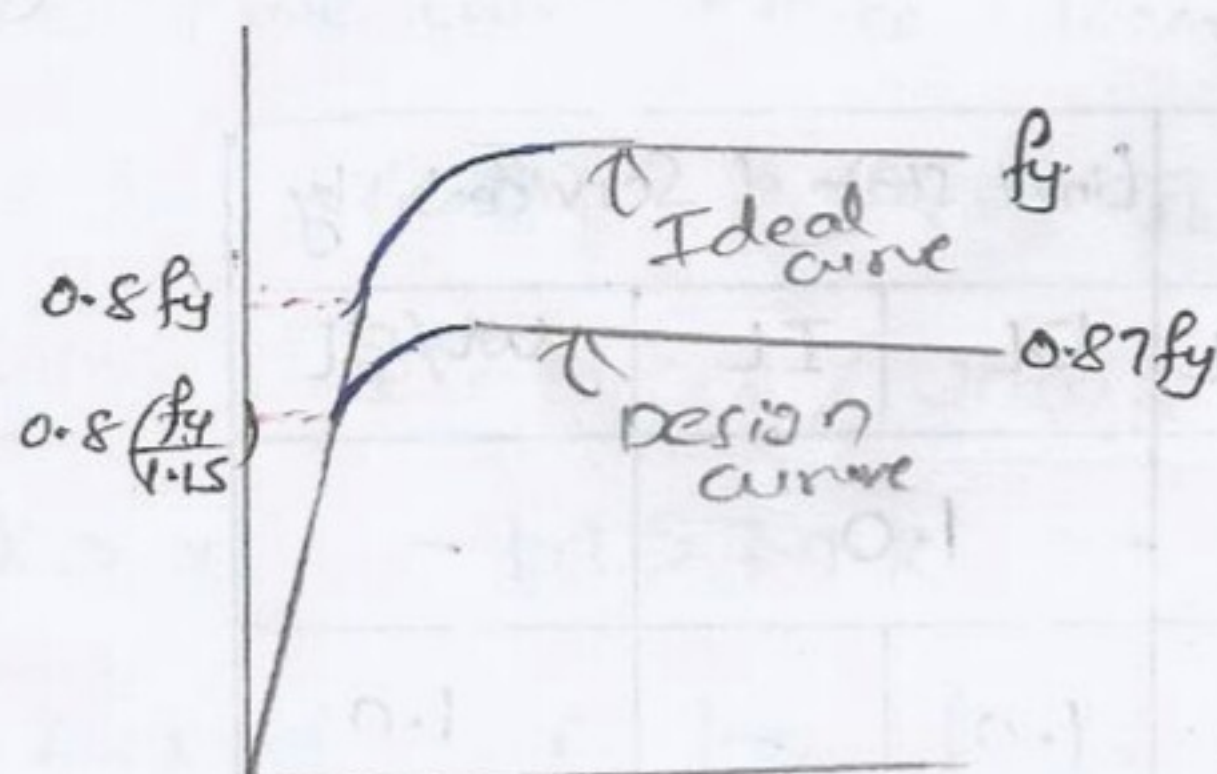
∴ Max Strain in Concrete $E_{cu} = 0.0035$



Stress-Strain Curve for Steel

a) For Mild steel (Fe 250)

It has definite yield point

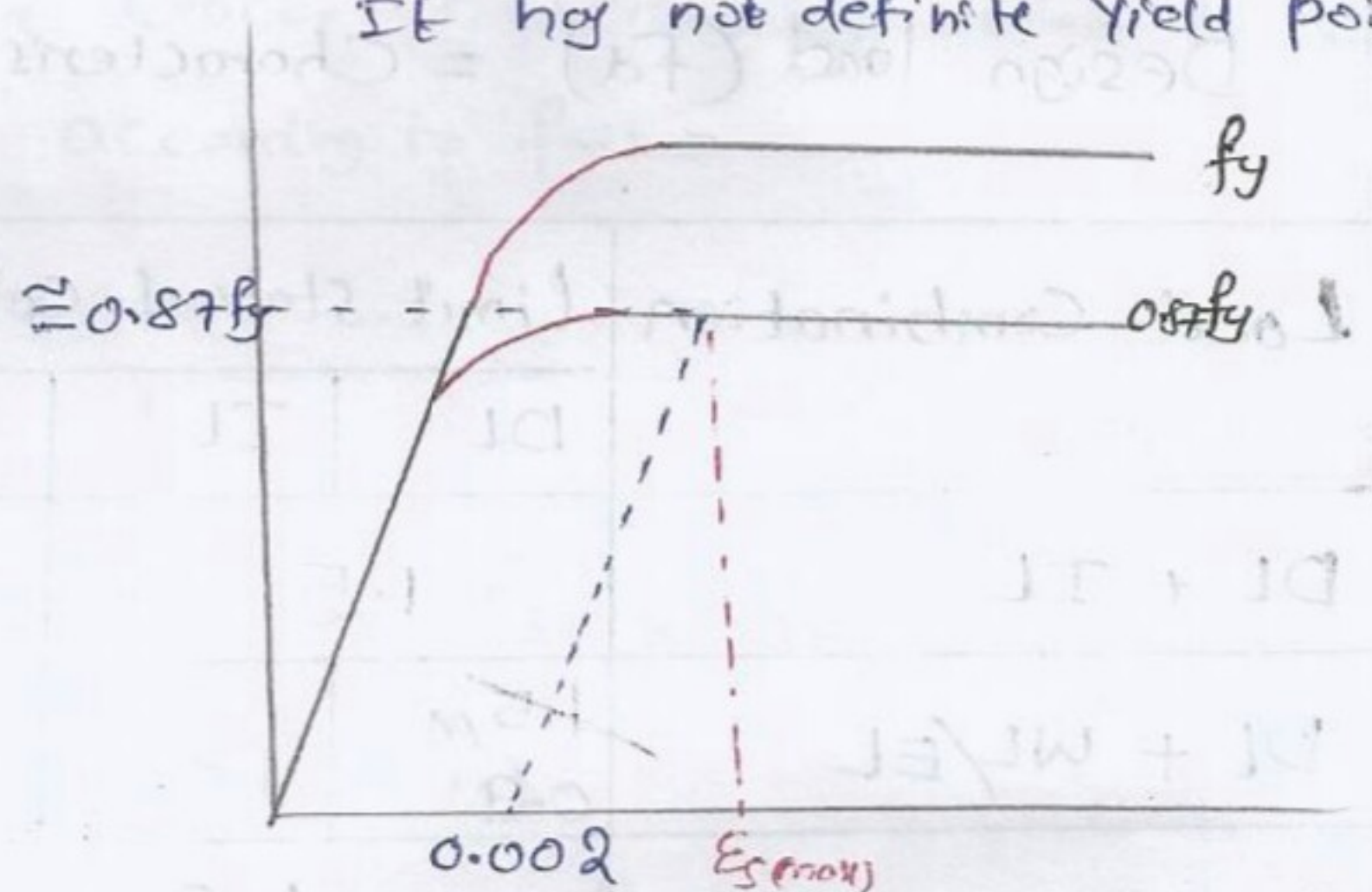


$$\text{Max Strain}(E_s) = \frac{0.87f_y}{E_s}$$

→ Curve is 'Linear' up to
 $0.8f_y$ & $0.8 \times (0.87f_y)$ for
Ideal & design curve
respectively. (for all type of steel)

b) For Cold worked deformed bars
(Fe 415 & Fe 500)

It has not definite yield point



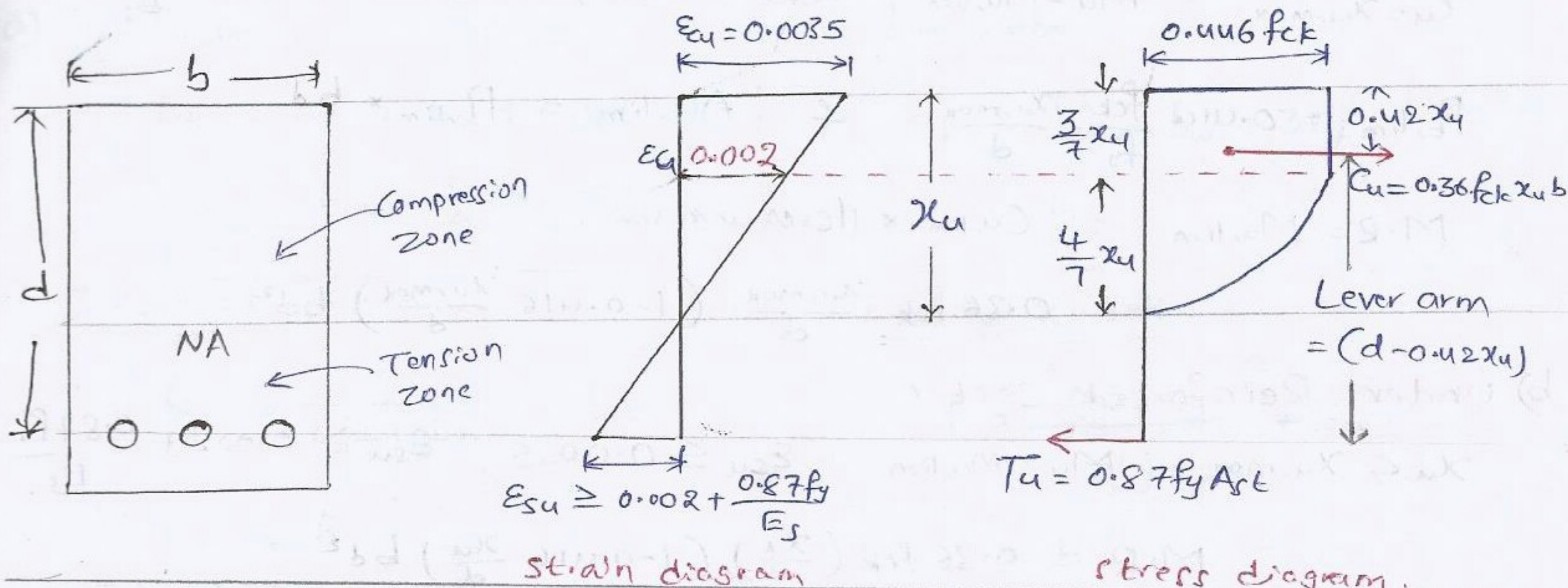
$$\text{Max Strain}(E_{su}) \geq 0.002 + \frac{0.87f_y}{E_s}$$

Characteristic yield strength

(i.e. Yield strain at failure)

$$\begin{aligned} &\approx 0.2\% \text{ of proof stress} \\ &\approx 0.87f_y \end{aligned}$$

Singly Reinforced Section



Strain diagram

Stress diagram

$\frac{x_u}{d} = \frac{\epsilon_{cu}}{\epsilon_{cu} + \epsilon_{su}}$ ($\because \epsilon_{su} \geq 0.002 + \frac{0.87f_y}{E_s}$, Hence if ϵ_{sc} is not constant, $\therefore$ It is not possible to calculate (x_u/d) from this formula)

① $\frac{x_{u,max}}{d} = \frac{\epsilon_{cu}}{\epsilon_{cu} + (\epsilon_{su})_{min}} = \frac{0.0035}{0.0035 + (0.002 + \frac{0.87f_y}{E_s})} = \frac{700}{1100 + 0.87f_y}$

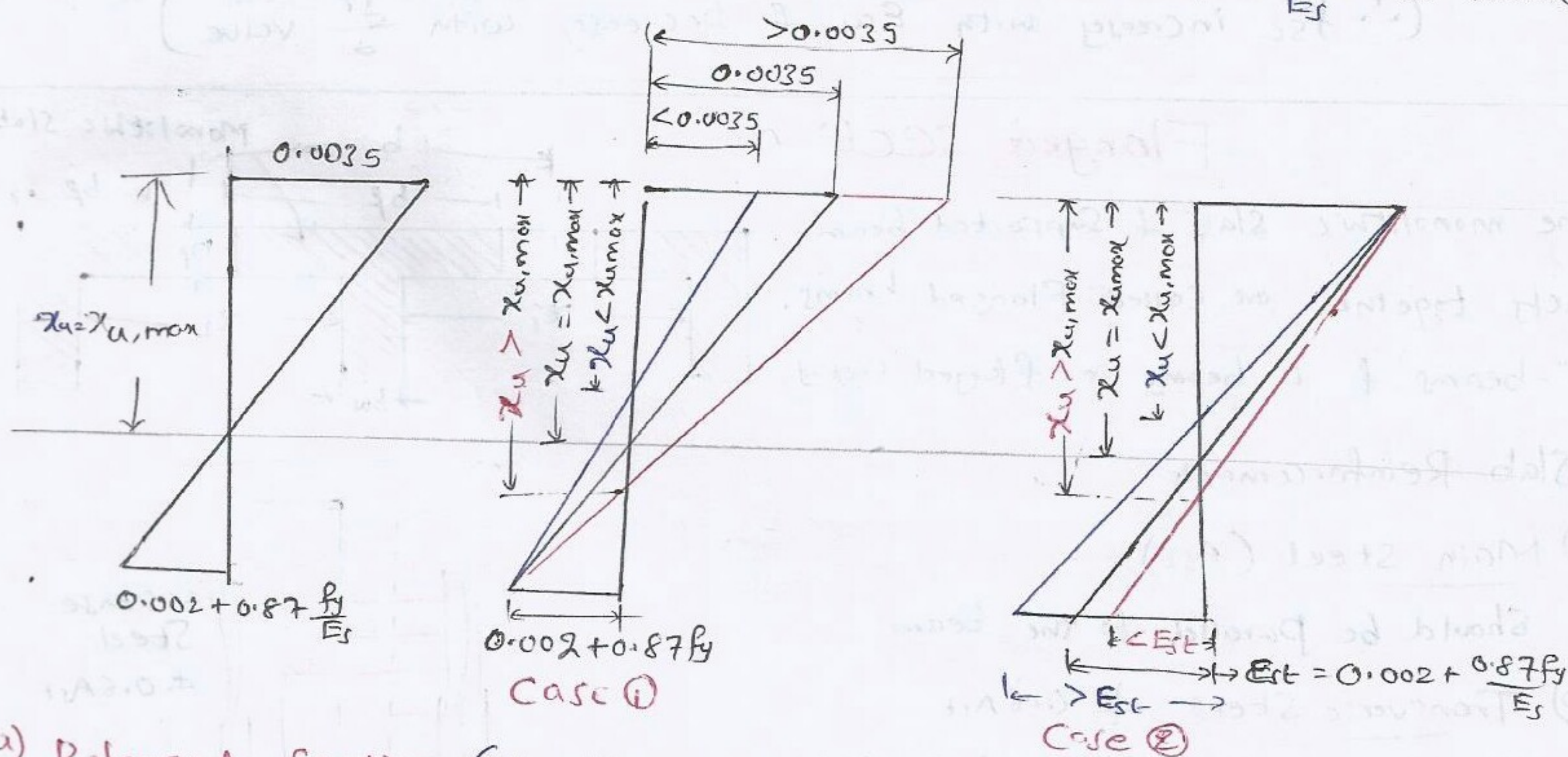
f_y	250	415	500
$\frac{x_{u,max}}{d}$	0.53 (0.521)	0.48 (0.479)	0.46 (0.456)

② $\frac{x_u}{d} = \frac{0.87f_y}{0.36f_{ck}} \left(\frac{A_{st}}{bd} \right) = 2.117 P_t \cdot \frac{f_y}{f_{ck}}$ ($\because P_t = \% \text{ of steel} = \frac{A_{st}}{bd}$)

In analysis, we had made two important assumptions that

Assumption 1 - Max Compressive Strain in Concrete = 0.0035 at failure

Assumption 2 - Min Tensile Strain in Steel Reinforce = $0.002 + \frac{0.87f_y}{E_s}$ at failure.



- Balanced Section ($x_u = x_{u,max}$) $\rightarrow$ Both assumptions Satisfied
- Under-Reinforced section ($x_u < x_{u,max}$) $\rightarrow$ Both assumptions are Satisfied in both cases.
- Over-Reinforced Section ($x_u > x_{u,max}$)
 - Case (1) $\rightarrow$ It violates the assumption ② although assumption ① Satisfied
 - Case (2) $\rightarrow$ It violates assumption ① although assumption ② Satisfied.

a) Balanced Section

$$x_u = x_{u, \max} \quad M_u = M_{u, \lim} \quad \epsilon_{cu} = 0.0035, \quad \epsilon_{su} = 0.002 + \frac{0.87 f_y}{E_s}$$

$$P_{t, \lim} = 0.416 \frac{f_{ck}}{f_y} \frac{x_{u, \max}}{d} \quad \& \quad A_{st, \lim} = P_{t, \lim} \times b d$$

$$M \cdot R = M_{u, \lim} = C_{u, \lim} \times (\text{lever arm})_{\lim}$$

$$= 0.36 f_{ck} \frac{x_{u, \max}}{d} \left(1 - 0.416 \frac{x_{u, \max}}{d}\right) b d^2$$

b) Under Reinforced section

$$x_u < x_{u, \max} \quad M_u < M_{u, \lim} \quad \epsilon_{cu} \leq 0.0035, \quad \epsilon_{su} \geq 0.002 + \frac{0.87 f_y}{E_s}$$

$$M \cdot R = 0.36 f_{ck} \left(\frac{x_u}{d}\right) \left(1 - 0.416 \frac{x_u}{d}\right) b d^2$$

It Satisfies all requirements, hence it is safe & economical

c) Over Reinforced section

$$x_u > x_{u, \max} \quad M_u < M_{u, \lim} \quad \epsilon_{cu} \geq 0.0035, \quad \epsilon_{su} \leq 0.002 + \frac{0.87 f_y}{E_s}$$

It violates at least one requirement in any case.

Hence it is not safe. And is better to redesign (by increasing d)

(or) It is better to go for Doubly Reinforced Sections.

Doubly Reinforced Sections

Adopted when ($M_u > M_{u, \lim}$) & O.R.S of Singly Reinf occur

$$\text{Strain in Compression reinforcement } \epsilon_{su} = \frac{0.0035 (x_u - d')}{x_u}$$

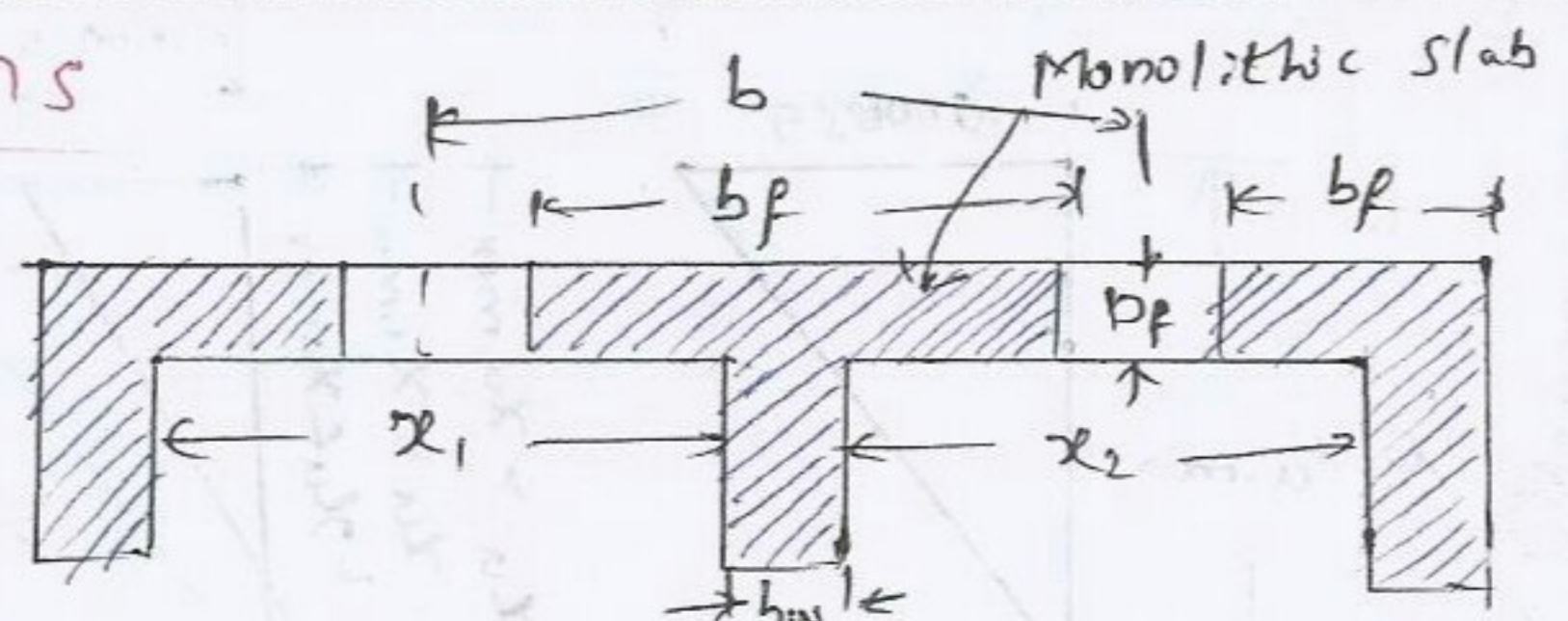
f_{sc} = Design stress in Compression reinforcement,

→ Depends on ϵ_{su} value from Stress-strain curve

($\therefore f_{sc}$ increases with ϵ_{su} & decreases with $\frac{d'}{d}$ value)

Flanged sections

The monolithic slab & supported beam act together are called Flanged beams. T-beams & L-beams are flanged sections.



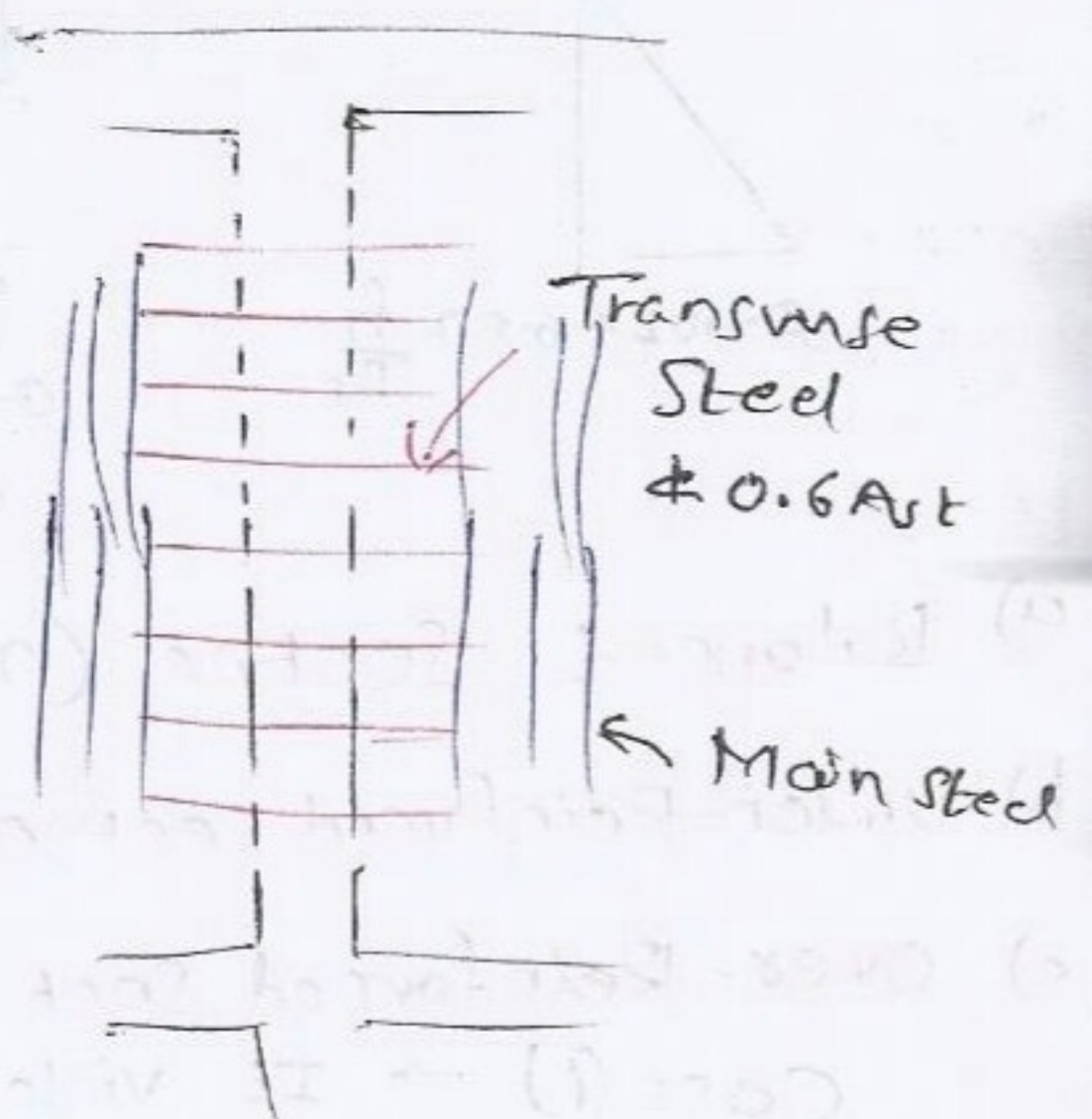
Slab Reinforcement

a) Main steel (A_{st})

Should be parallel to the beam

b) Transverse Steel $\& 0.6 A_{st}$

Provided at the mid span of the slab



Calculation of Effective flange width (which ever is less)

a) T-beam

$$b_f = \frac{L_o}{6} + b_w + 6D_f$$
$$= b_w + \frac{1}{2}(x_1 + x_2)$$

Isolated T-beam

$$b_f = \frac{L_o}{\left(\frac{L_o}{b}\right) + 4} + b_w$$
$$= b$$

b) L-beam

$$b_f = \frac{L_o}{12} + b_w + 3D_f$$
$$= b$$

Isolated L-beam

$$b_f = \frac{0.5L_o}{\left(\frac{L_o}{b}\right) + 4} + b_w$$
$$= b$$

Where

L_o = Distance b/n the points of zero moments in the beam (Simply supported)
 $= 0.7L_{eff}$ (for Continuous beam)

Effective thickness of Flange (y_f)

$$y_f = 0.15x_{u,max} + 0.65D_f \geq D_f$$

Conditions

① Compressive stress in the flange ($0.446f_{ck}$) will be uniform, When

$$D_f \leq \frac{3}{7}x_u \leq 0.43x_u \quad \text{or} \quad \text{(Under Reinforced)}$$

$$\leq \frac{3}{7}x_{u,max} = \frac{3}{7}x (\approx 0.48d) \leq 0.2d \quad \text{(Balanced section)}$$

② if $x_u < D_f$ & $b_f \approx b_w$

Then T-beam behaves like a Rectangular beam.

Limit State of Collapse in Shear

Shear stresses in the beam are caused by the variation of B.M along the span. (SF = Constant, is called Shear span)

Failure modes due to Shear

Depending upon the ratio of B.M(M) + S.F(V) at different section, there may be three regions of cracks in the beam.

a) Web Shear

→ Cracks developed near the web & they are inclined ($\approx 45^\circ$)

b) Flexural Tension Shear

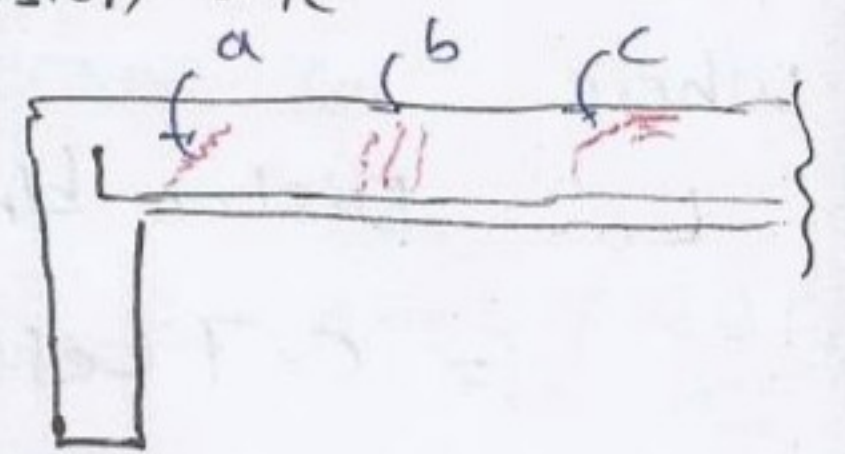
→ Occurs when tensile stresses reaching permissible stresses f ($M \gg V$)

→ They are vertical in nature, starting from tension zone

c) Flexural Compressive Shear

→ Occurs when large shear and crushing of concrete

→ They are vertical & inclined, observe in comp zone.



Design of Shear Stress

a) Nominal Shear Stress (τ_v) = $\frac{V_u}{bd}$ (Uniform depth)

= $\frac{V_u \pm \frac{M_u}{d} \tan \phi}{bd}$ (Varying depth)

b) Design shear strength (τ_c)

Depends on f_{ck} & A_t $\left[\begin{array}{l} \tau_c \approx 0.25 \sqrt{f_{ck}} \text{ (L.S.M)} \\ 0.16 \sqrt{f_{ck}} \text{ (W.S.M)} \end{array} \right]$

(+ve for $M \uparrow$ & $d \uparrow$)
& vice versa

c) Maximum design Shear Stress ($\tau_{c,max}$) of Concrete with Shear Reinforcement

Grade of concrete	M15	M20	M25	M30	M35	M40
$\tau_{c,max}$	2.5	2.8	3.1	3.5	3.7	4.0
	2.0	2.8	3.2	3.6	4.0	4.4

(L.S.M)
(W.S.M)

Case (i) ($\tau_v < \tau_c$)

Provide minimum shear reinforcement, in the form of stirrups.

$$\frac{A_{sv}}{b s_v} \geq \frac{0.4}{0.87 f_y}$$

A_{sv} = Shear Reinforcement area
 s_v = Spacing of stirrups

Case (ii) ($\tau_c < \tau_v < \tau_{c,max}$)

Extra shear force ($V_{us} = (\tau_v - \tau_c) bd$) shall be provided.

Spacing of stirrups can be calculated by

a) For Vertical stirrups

$$V_{us} = \frac{0.87 f_y A_{sv} d}{s_v}$$

b) Inclined stirrups

$$V_{us} = \frac{0.87 f_y A_{sv} d}{s_v} (\sin \alpha + \cos \alpha)$$

c) Bent-up bars

$$V_{us} = 0.87 f_y A_{sv} \sin \alpha$$

Note:

$\alpha \neq 45^\circ$ & $f_y \neq 415 \text{ MPa}$

$s_v \neq \begin{cases} 0.75d \\ 300 \text{ mm} \end{cases}$ whichever is less

→ Minimum shear reinforcement, provided to resist the "Principal Tension"

→ Bent-up bars are provided in a beam to resist the "Diagonal Tension".

Limit State of Collapse in Bond

Bond - The adhesive force between concrete & Reinforcement

Bond stress (τ_{bd}) - The intensity of this adhesive force is called Bond stress

→ Is the shear stress acting parallel to the reinforcing bar on the interface b/w the bar and the concrete.

Grade of Concrete	M20	M25	M30	M35	$\geq M40$
τ_{bd} for plain bars in Tension	1.2	1.4	1.5	1.7	1.9

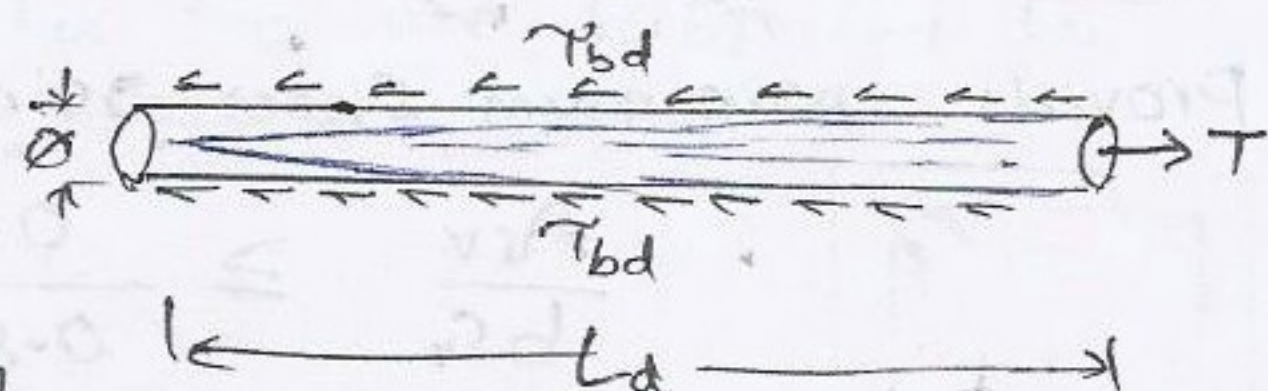
→ Increased by 60% for deformed bars $= 1.6 \tau_{bd}$

→ Increased by 25% when bars in Compression, $= 1.25 \tau_{bd}$

Development Length

Due to the bond, the total length of the bar may not be effected when it is stressed.

L_d = The minimum length of the bar effected from Yield stress to the Zero stress



∴ Development Length (when bar in Tension)

$$L_d = \frac{\phi \sigma_s}{4 \tau_{bd}} = \frac{0.87 f_y \phi}{4 \tau_{bd}}$$

$$\left[\begin{array}{l} \frac{\pi \phi^2 \sigma_s}{4} = (\pi \phi L_d) \tau_{bd} \\ M_1 \text{ for } 0.87 f_y \\ L_0 = 12\phi \} \text{ greater} \\ = d \end{array} \right.$$

$$L_d \geq \frac{M_1}{V} + L_0 \quad (\text{Normal Condition})$$

$$\geq 1.3 \frac{M_1}{V} + L_0 \quad (\text{Ends of bars are Confined by compressive reaction})$$

* When Bars in bundle: ($\phi \neq 32\text{mm}$)

L_d of each bar shall be increased by 10%, 20%, & 33% for 2 bars, 3 bars & 4 bars in contact respectively.

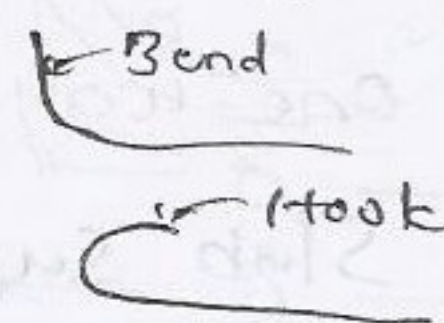
Anchoring Bars (Extra length for safety)

a) In Tension

(i) For deformed bars, may not require anchorage (∵ provided L_d is sufficient)

(ii) Bend Anchorage value $= 4\phi \times \left(\frac{\text{Angle of bend}}{45^\circ} \right)$

(iii) Hooks Anchorage value $= 16\phi$ (U-type hooks)



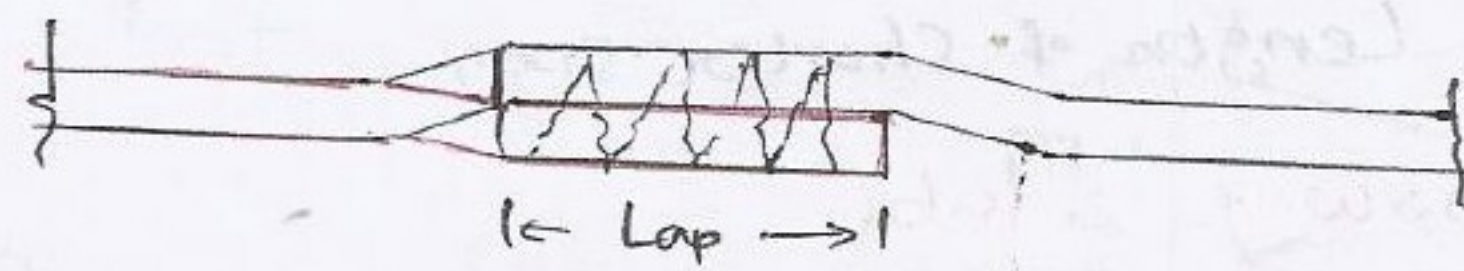
b) In Compression

Anchorage value $= L_d$ (for any type of bend & hook)

Reinforcement Splicing

→ Splicing should not be done when

$B.M > 0.5 M_{1R}$ & No. of Splicing bars $< 50\%$ & $\phi > 36\text{mm}$



Lap length

a) In Bending tension $= L_d$ } greater $(\phi = \text{least dia})$
 $= 30\phi$

b) In direct Tension $= 2L_d$ } greater
 $= 30\phi$

c) In Compression $= L_d$ } greater
 $= 24\phi$

Limit state of Collapse in Torsion

Torsion effect is majorly seen in eg: L-beams, Space frame, etc.
Hence stirrups are provided to resist the torsion.
called Torsion Reinforcement also called Transverse/Lateral reinforcement. These are $\perp$ to the Main/Longitudinal reinforcement.

$$\text{Equivalent nominal shear stress } (\tau_{ve}) = \frac{V_e}{bd}$$

$$\text{where } (V_e = V_u + 1.6 \frac{T_u}{b})$$

T_u = Actual torsion moment.

Case(i) ($\tau_{ve} < \tau_c$)

Provide minimum shear reinforcement in the form of stirrups

$$\frac{A_{sv}}{b s_v} \geq \frac{0.4}{0.87 f_y}$$

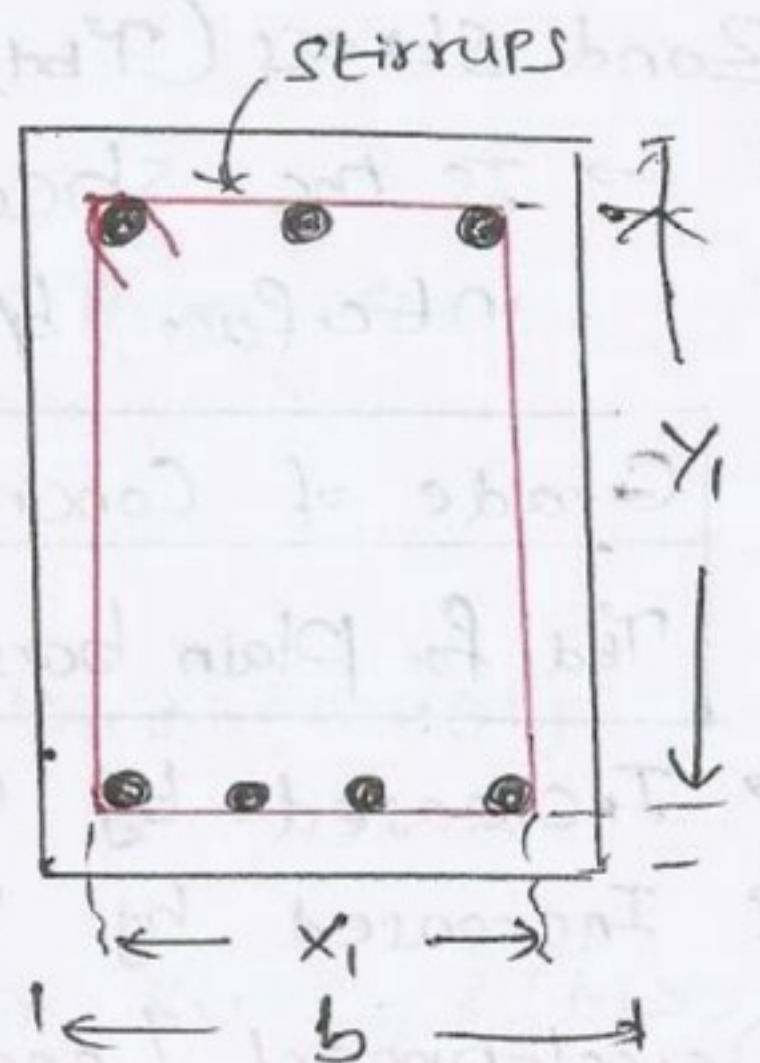
Case(ii) ($\tau_{ve} > \tau_c$)

Both Longitudinal & transverse reinforcement shall be provided as

$$\frac{A_{sv}}{b s_v} \geq \frac{(\tau_{ve} - \tau_c)}{0.87 f_y}$$

$$\text{Spacing of stirrups } \left\{ \begin{array}{l} x_1 \\ \frac{x_1 + y_1}{4} \\ 300 \text{ mm} \end{array} \right\} \text{ Least.}$$

~~Beam is subjected to torsion
when total depth of beam = 450-750mm~~



Slabs

a) One-way Slabs

Slab supported by one-directional beams

Adopted when

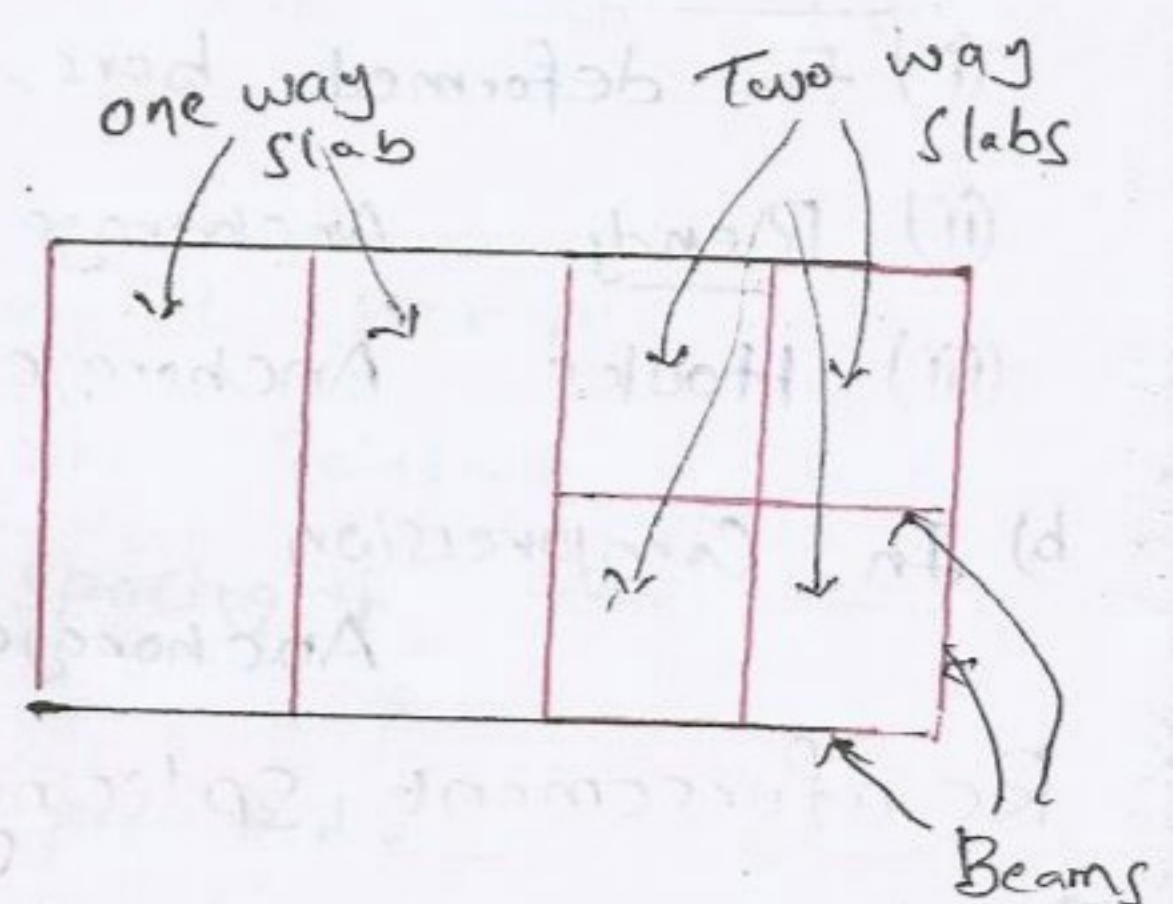
$$\frac{\text{Length of longer span}}{\text{Length of shorter span}} \geq 2$$

b) Two-way Slabs

Slab supported by all four edges & sides of beams

Adopted when

$$\frac{\text{Length of longer span}}{\text{Length of shorter span}} < 2$$



c) Flat Slabs

The slabs should be supported by columns only.
(i.e. There is no beams in the structure)

Generally seen in Multi-complex, Theatre, etc.

Min thickness of Flat slab
= 125 mm
Column head vertex
angle = 90°

Limit State of Collapse in Compression (Columns)

a) Wall $D > 4d$

b) Pedestal ($L_{eff} \leq 3b$), ($D \leq 3b$), ($b < D$)

c) Column ($L_{eff} > 3b$)

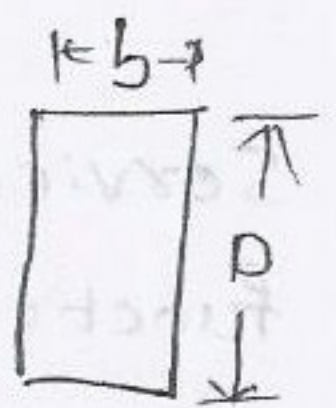
And also satisfy the following requirements.

$$L \leq 60b \quad (\text{Both ends restrained})$$

$$\leq \frac{100b^2}{D} \quad (\text{Cantilever Column})$$

$$D \leq 4b \quad \text{f}$$

$$b < D$$



b = least lateral dimension
 D = Great lateral dimension

→ Columns used in buildings when they are subjected to Axial force only

→ Columns used in Framed Structures, when they are subjected to Both Combined bending & axial force.

1) Tied Column

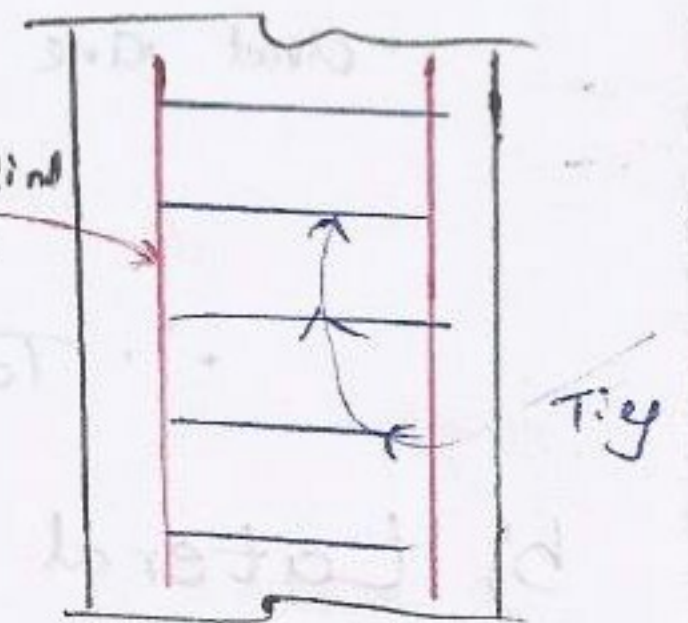
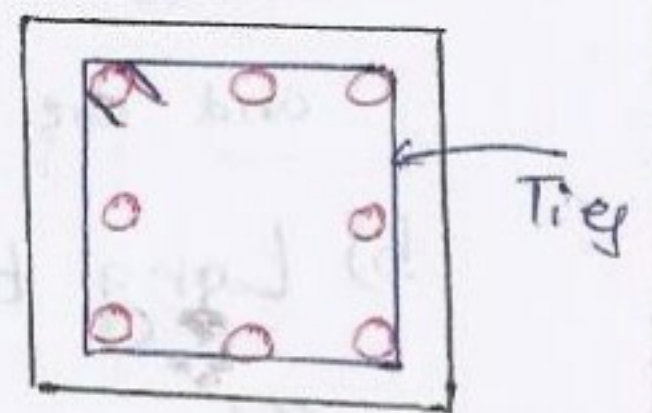
Square/Rectangular columns are mostly of this type

2) Column with helical Reinforcement

Circular and octagonal columns are mostly of this type

$$\text{Strength of (helical column)} = 1.05 \times (\text{Strength of tied column})$$

$$\text{if } \left(\frac{\text{Volume of spiral Reinf}}{\text{Volume of core}} \right) \geq 0.36 \left(\frac{A_g}{A_{core}} - 1 \right) \frac{f_{ck}}{f_y}$$



Longitudinal Reinforcement (A_{sc})

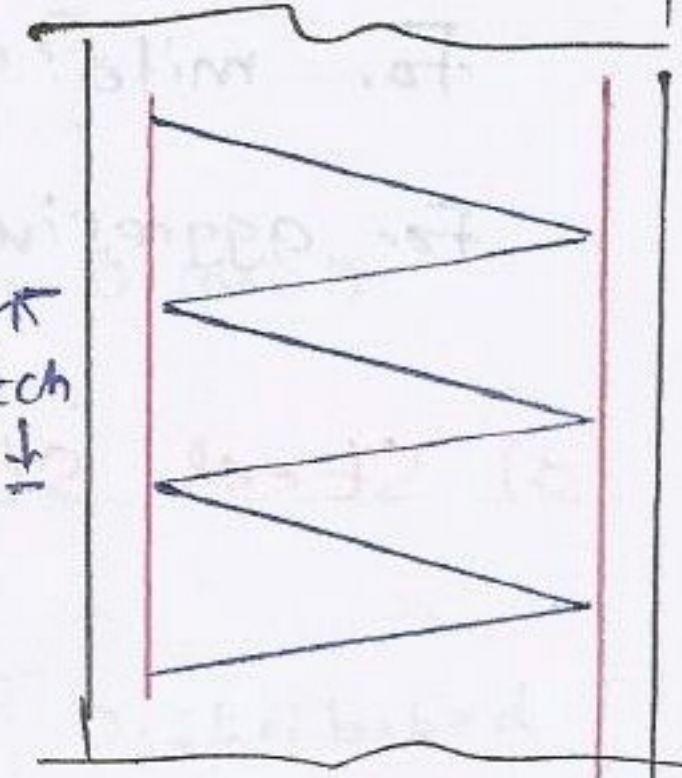
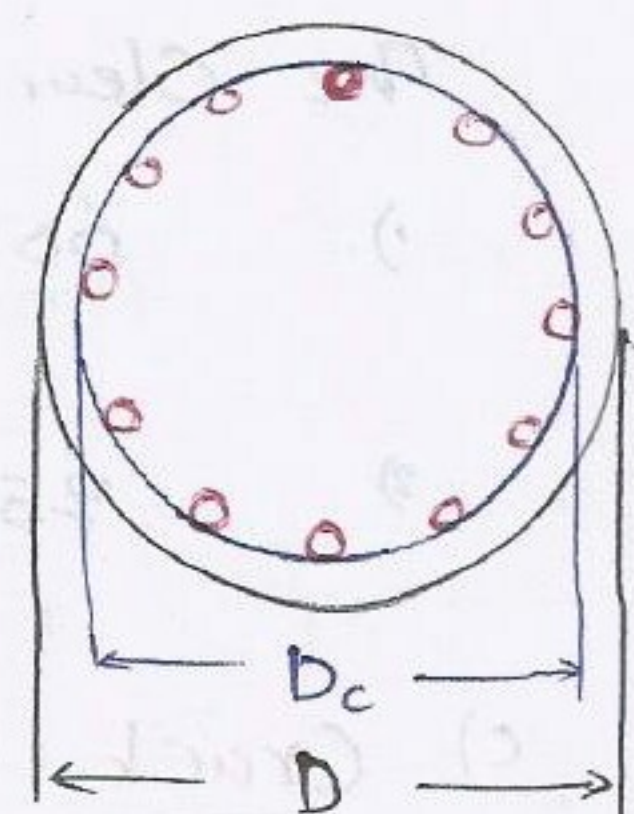
1) $A_{sc} \nlessgtr 0.8\%$ of A_g (Columns) $\nlessgtr 0.15\%$ of A_g (Pedestal)
 $\nlessgtr 4\%$ of A_g

2) No. of longitudinal bars $\nlessgtr 4$ (Rectangular Tied Columns)
 $\nlessgtr 6$ (Circular helical Column)

3) Dia. of longitu. bar $\nlessgtr 12\text{ mm}$
Spacing of " " $\nlessgtr 300\text{ mm}$

4) Pitch/spacing of bars $\nlessgtr \left. \begin{matrix} b \\ 16\phi \\ 300 \end{matrix} \right\} \text{least}$

Tied Column



Helical Reinf
(D_c = core dia)

Helical Reinforcement

1) Minimum pitch $\nlessgtr \left. \begin{matrix} 25\text{ mm} \\ 3\phi \end{matrix} \right\} \text{least}$

2) Maximum pitch $\nlessgtr \left. \begin{matrix} 75\text{ mm} \\ \frac{1}{6} D_c \end{matrix} \right\} \text{least}$

* Minimum eccentricity

$$e_{min} \nlessgtr \left[\frac{L}{500} + \frac{b \text{ or } D}{30} \right] \text{ least}$$

20 mm

* Ultimate load carrying capacity of Column (P_u) = $0.4f_{ck}A_c + 0.67f_yA_{sc}$

* Slenderness Ratio (λ) = $\frac{\text{Effective length}}{\text{least lateral dimension}} = \frac{L_{eff}}{b} \leq 12$ (Short Column)
 > 12 (Slender Column)

Limit State of Serviceability

Serviceability is the fitness of the structure to serve the desired function satisfactorily under service load.

a) Deflection

Final Deflection $\nless \frac{\text{Span}}{250}$ (before casting)

$\nless \frac{\text{Span}}{250}$ } least (After construction)
20 mm

a) Short term Deflection

These occurs immediately after the application of load.

and are calculated on the basis of short term elastic modulus $E_c = 5000 \sqrt{f_{ck}}$

b) Long term Deflection

These occurs due to Creep and shrinkage effect, under sustained load.

and are calculated on the basis of modified elastic modulus $(E_{ce}) = \frac{E_c}{1+\theta}$ (θ = Creep Coefficient)

$\therefore$ Total Deflection = Short + Long term deflection

b) Lateral Stability

When the width of compression face is small as compared to the depth of the beam, it may fail in buckling.

The clear distance between lateral restraints should be lesser of following

1) $60b$ (or) $\frac{250b^2}{D}$ (For SS + Continuous beam)

2) $25b$ (or) $\frac{100b^2}{D}$ (For Cantilever beam)

c) Crack width

For mild environment, Surface Crack $\nless 0.3 \text{ mm}$

For aggressive environment, Surface Crack width $\nless 0.004 \times \text{Nominal Cover}$

d) Steel stress of service loads (f_s)

$$f_s = 0.58 f_y \times \frac{(A_{st})_{\text{required}}}{(A_{st})_{\text{provided}}}$$

Codal Provisions

Spacing of Reinforcement $(S) = \frac{A_{\phi}}{A_{st}} \times 1000$

a) Beams

Horizontal spacing $\nless Max Dia of bar \left. \begin{array}{l} \\ Max agg + 5 mm \end{array} \right\} \text{greater}$

Vertical spacing $\nless Max Dia of bar \left. \begin{array}{l} \frac{2}{3} \text{ of Max agg} \\ 15 mm \end{array} \right\} \text{greater}$

b) Side face Reinforcement

$\nless Web thickness \left. \begin{array}{l} \\ 300 mm \end{array} \right\} \text{Least}$

c) Shear Reinforcement

$\nless 0.75d \left. \begin{array}{l} \\ 300 mm \end{array} \right\} \text{Least}$

d) Slabs

Normal condition

$\nless 3d \left. \begin{array}{l} \\ 300 mm \end{array} \right\} \text{Least}$

against shrinkage & Temperature

$\nless 5d \left. \begin{array}{l} \\ 450 mm \end{array} \right\} \text{Least}$

e) Columns

Longitudinal pitch

$\nless \text{Least lateral dimension} \left. \begin{array}{l} \\ 16\phi \\ 300 mm \end{array} \right\} \text{Least}$

Reinforcement Details

a) Beams

$$(A_{st})_{min} = \frac{0.85}{f_y} b d$$

$$(A_{sc})_{min} = 0$$

$$(A_{st})_{max} = 4\% \text{ of gross area} \\ = 0.04 b d$$

$$(A_{sc})_{max} = 0.04 b d$$

b) Side face Reinforcement

provided when depth of web $> 750 mm$

Total area of such Reinforcement $\nless 0.1\%$ of web area

Distributed equally along two faces

c) Columns

$$(A_{sc})_{min} = 0.8\% \text{ of } b d$$

$$(A_{st})_{max} = 4\% \text{ of } b d \text{ (Normal Condition)} \\ = 6\% \text{ of } b d \text{ (may involve practical difficulties in placing \& compacting)}$$

d) Slabs

$$(A_{st})_{min} = 0.15\% \text{ of c/s area (Mild steel)} \\ = 0.12\% \text{ of c/s area (Deformed or welded)}$$

Size of bar (ϕ)

- $\phi \nless 12\text{ mm}$ (Columns)
- $\nless 36\text{ mm}$ (Lap splices)
- $\nless \frac{D}{8}$ (slabs)

Although Dia of bars available in market from 5-50 mm

Effective lengths for beams (For Columns Same as Rolled Steel Sections)

a) Simply Supported

$$L_{\text{eff}} = L + \frac{d}{c/c} \text{ } \left. \vphantom{L_{\text{eff}}} \right\} \text{Least}$$

b) Cantilever

$$L_{\text{eff}} = L + \frac{d}{2}$$

c) Continuous

→ Case (i) if $\left[b < \frac{L}{12} \right]$

$$L_{\text{eff}} = L + \frac{d}{c/c} \text{ } \left. \vphantom{L_{\text{eff}}} \right\} \text{Least}$$

→ Case (ii) if $\left[b \geq \frac{L}{12} \right]$ $\left. \vphantom{L_{\text{eff}}} \right\} \text{Least}$
600 mm

C/c = Centre to Centre of Supports
L = clear span

a) $L_{\text{eff}} = L$

(For starting end fixed)

b) $L_{\text{eff}} = \left. \begin{matrix} L + \frac{d}{2} \\ L + \frac{b}{2} \end{matrix} \right\} \text{Least}$

(For starting end free)

d) Frames

$$L_{\text{eff}} = C/c$$

Effective Depths

	Beams (Mild & HYSD)			Slabs (Mild)	
	Span $\leq 10\text{ m}$	Span $> 10\text{ m}$	T-beams	one way	Two-way (Span $\leq 3.5\text{ m}$)
	$\frac{\text{Span}}{d} \nless$	$\frac{10}{d} \nless$	$\frac{\text{Span}}{d}$	$\frac{\text{Span}}{D} \nless$	$\frac{\text{Span}}{D} \nless$
Cantilever	7	-	10-12	7	-
Simply Supported	20	20	12-15	20	35
Continuous	26	26	-	26	40

Note:

→ For Slabs of HYSD bars, the above value should be multiplied by 0.8

→ This Effective depth shall be used to know the Vertical deflection limit.

Nominal Cover or Clear Cover

	Nominal Cover ⌀
Beams & Slabs	⌀
Columns	40 mm
Footings	50 mm

Exposure Condition	Nominal Cover ⌀	Minimum Concrete
Mild	20 mm	M20
Moderate	30 mm	M25
Severe	45 mm	M30
Very Severe	50 mm	M35
Extreme	75 mm	M40

Note:

→ For Columns of $b \leq 200 \text{ mm}$ & $\phi \leq 12 \text{ mm}$, then Nominal Cover = 25 mm

→ For Mild Exposure Condition & $\phi \leq 12 \text{ mm}$, then NC reduced by 5 mm

For Severe & very severe & $f_{ck} \geq M35$, then NC reduced by 5 mm.

Age Factor

Generally Compressive Strength of concrete taken at 28 days of f_{ck} of 150 mm cube

Setting Time	Age factor
7 days	0.7
28 days	1.0
3 months	1.10
6 months	1.15
1 year	1.20

eg:

Strength of M20 grade
Concrete after 6 months
 $= 20 \times 1.15$
 $= 23 \text{ N/mm}^2$

Deflection recovery

→ If imposed loads are removed within 24 hrs, The structure should be recover its deflection at least 75%, if not test repeat after 72 hrs

→ Again after lapse of 72 hrs, if recovery is less than $< 80\%$, the structure shall not be acceptable.

→ If max deflection (mm) during 24h under load is $< \frac{4\alpha L^2}{D}$
($\because L = \text{Effective span (m)}, D = \text{overall depth (mm)}$)